

Contents

1. INTRODUCTION	1
1.1 Background of the Study	1
1.2 Problem Statement	1
1.3 Aim of the Project	1
1.4 Objectives of the Study	2
1.5 Significance of the Project	2
1.6 Scope of the Project.....	2
1.7 Project Challenges.....	2
1.8 Contribution of the Study	3
2. LITERATURE REVIEW	3
2.1 Existing Projects and Their Features.....	3
2.2 Limitations of Existing Projects	4
2.3 Justification for the Present Study.....	5
3. METHODOLOGY	5
3.1 Components.....	5
3.2 Interaction Between Components	6
3.3 Protocols and Network Services	7
3.3 Process Flow and Smart Integration.....	8
4. IMPLEMENTATION	9
4.1 Network Topology Overview.....	9
4.2 IoT Device Integration	18
5. RESULTS & DISCUSSION.....	19
5.1 Results	19
5.2 Discussion (Limitations & Applications).....	19
6. CONCLUSION.....	20
7. FUTURE ENHANCEMENT.....	21
8. REFERENCE.....	21

Hotel Network Design

1. INTRODUCTION

1.1 Background of the Study

This project addresses the growing need for a secure, scalable, and intelligent network infrastructure within modern hotels. As hospitality environments become increasingly digital, hotels depend on interconnected ICT systems to support guest services, administrative operations, security, and smart-room automation. The rise of devices such as smartphones, laptops, IoT sensors, surveillance cameras, and automated access systems has made reliable network performance essential across all hotel floors and departments.

To meet these demands, this project designs a complete hotel network management system using Cisco Packet Tracer, modelling a multi-floor hotel with dedicated VLANs, dynamic OSPF routing, automated DHCP and DNS services, and centralized web and email servers. The integration of IoT devices further enhances the system by enabling smart environmental control, motion-based automation, RFID access, and automated fire-response alerts. By combining traditional networking with intelligent automation, the project demonstrates how modern hotels can improve guest experience, enhance operational efficiency, and strengthen security through a well-engineered ICT infrastructure.

1.2 Problem Statement

Many existing hotel networks lack the capacity or structure to support current digital demands. Fragmented architectures, poor segmentation, outdated routing practices, and limited security controls create environments where congestion, inefficiency, and cybersecurity risks are common. Hotels often attempt to serve guests, staff, and internal systems through the same unsegmented network, leading to slow performance and exposure of sensitive data. The absence of IoT integration also leaves many facilities without automation for lighting, temperature control, access management, or emergency response. These shortcomings highlight the need for a unified and intelligent network design capable of addressing modern hospitality requirements.

1.3 Aim of the Project

The aim of this project is to design and simulate a fully functional hotel network system using Cisco Packet Tracer that demonstrates how traditional networking principles and modern IoT technologies can work together to support hotel operations. The goal is to construct an infrastructure that not only delivers high-speed connectivity and secure communication but also enables smart automation within guest rooms and service departments. Through this project, the hotel network is envisioned as an integrated digital ecosystem rather than a collection of isolated devices.

1.4 Objectives of the Study

The primary objective is to develop a network that reflects the real-world operational demands of a multi-floor hotel environment. This includes the structured implementation of VLANs for traffic segmentation, OSPF for scalable routing, DHCP and DNS for automated addressing and name resolution, and centralized servers for internal services. Another objective is to incorporate IoT devices such as smart lights, thermostats, cameras, and fire sensors to demonstrate how automation can improve guest comfort, enhance safety, and reduce operational costs. By validating these components through simulation, the project seeks to demonstrate the feasibility of intelligent hotel infrastructures.

1.5 Significance of the Project

This project holds significant value for both academic study and practical application. Academically, it provides a comprehensive example of how complex networking concepts are applied in a real-world scenario, bringing together routing, switching, server management, and IoT integration in a single coherent environment. From a practical standpoint, the design models the direction in which modern hotels are moving toward smart automation, efficient resource use, and secure digital systems. The completed simulation serves as a proof of concept that hotels can adopt technologically advanced infrastructures without sacrificing reliability or security.

1.6 Scope of the Project

The scope of this project encompasses the design and simulation of the entire digital infrastructure of a hotel. This includes three guest floors, a reception area, an accounts department, and a central server room, all connected through a hierarchical network. The system integrates multiple server-based services, wireless access points, and a range of IoT devices configured to interact through automation rules. The simulation focuses on demonstrating functionality, network performance, and intelligent behavior rather than real-world hardware deployment, although the architectural principles remain transferable to actual hotel environments.

1.7 Project Challenges

Designing a multi-floor hotel network presents several technical and operational challenges that must be carefully resolved. One major challenge involves ensuring secure segmentation between guest and staff networks, as unrestricted communication can expose sensitive administrative data to unauthorized access. Another difficulty lies in managing scalable routing across multiple routers and interconnected floors, requiring the correct implementation of OSPF to maintain stable and efficient communication paths. Integrating IoT devices adds its own set of challenges, as smart sensors and actuators rely on real-time responsiveness and consistent connectivity to perform automated functions. Configuring server-based services such as DHCP, DNS, web hosting, and email systems also requires precision to ensure service

reliability. These challenges collectively form a complex engineering problem that demands a carefully structured methodological approach and meticulous configuration.

1.8 Contribution of the Study

This study contributes both academically and practically to the understanding and implementation of advanced hotel networks. From an academic standpoint, it provides a detailed demonstration of how multiple networking concepts routing, switching, server operations, and IoT integration can be harmonized within a complex, multi-layered topology. It serves as a comprehensive learning model for students and engineers seeking to understand the interconnected nature of modern network designs. Practically, the project offers a realistic blueprint that hotels can adapt to upgrade their communication infrastructure, enhance guest experiences, and improve operational efficiency. The IoT automation framework presented in this study showcases how energy management, security monitoring, and emergency response can be improved through intelligent control systems. Ultimately, the project contributes a forward-thinking, technically sound model for the future of hotel network engineering.

2. LITERATURE REVIEW

2.1 Existing Projects and Their Features

Previous studies and simulation-based hotel network projects have explored a range of networking techniques, though often within limited scopes. The common features observed across existing works can be summarized as follows:

- **Basic VLAN Segmentation:** Many projects demonstrate the use of VLANs to separate traffic between guests and administrative staff. This segmentation helps reduce broadcast traffic and provides a basic layer of security. However, these implementations usually involve only one or two VLANs and seldom address multi-floor separation or department-specific subnetting.
- **Simple Routing Configurations:** Several academic projects implement static routing or single-router topologies. These designs offer an introductory understanding of routing but do not represent the complexity of hotel environments that require multi-router communication and dynamic route management. As a result, the routing aspect of existing studies remains limited in scope.
- **Minimal IoT Integration:** A number of simulations incorporate basic IoT devices such as motion sensors or door locks. While these examples illustrate the concept of automation, they do not extend into multi-device ecosystems where lighting, environmental control, security, and safety systems operate together as part of a smart-hotel framework.
- **Standard Server Implementations:** Some projects deploy DHCP, DNS, or Web Servers to demonstrate foundational enterprise services. These implementations show how servers

can support automated addressing or simple web hosting; however, they rarely include multi-page websites, email systems, or integrated IoT-based alerting capabilities.

- **Elementary Security Controls:** Existing research often introduces VLAN-based isolation but seldom implements Access Control Lists (ACLs), security policies for IoT devices, or measures that protect administrative traffic from guest networks. This results in designs that are functional for basic demonstrations but insufficiently secure for real-world hotel operations.

These existing works provide valuable foundational knowledge but do not reflect the full technological demands of modern hospitality networks. Their limitations open the way for more advanced and integrated models such as the hotel network presented in this project.

2.2 Limitations of Existing Projects

Despite offering useful insights, existing hotel network simulations exhibit several constraints that restrict their applicability in real-world contexts. The main limitations identified in the literature include the following:

- **Lack of Network Scale and Realism:** Many simulations operate on small topologies with minimal devices and single-floor structures. They fail to model multi-floor environments, interconnected departments, or the routing complexity required for large hotel buildings. This limits their relevance to modern hospitality infrastructures.
- **Insufficient IoT Automation:** Although some projects introduce IoT devices, they rarely demonstrate full automation workflows across different systems. Real hotels require coordinated automation involving motion detection, temperature control, access management, surveillance, and emergency response. Most existing simulations stop at basic demonstrations and do not capture the true potential of IoT integration.
- **Limited Server Functionality:** Existing works often exclude critical services such as email handling, advanced DNS configurations, or realistic web-hosting environments. Without these components, the simulated network fails to reflect how hotels manage bookings, communication, and internal workflows in real operation.
- **Weak Security Modelling:** VLAN segmentation is usually implemented only at a conceptual level, without comprehensive ACL enforcement or isolation of IoT devices. This can leave sensitive administrative systems exposed in real deployments. Security is therefore treated as a minor component rather than a central requirement.
- **Absence of Multi-Router Routing Protocols:** Many projects rely solely on static routing or small-scale networks, avoiding dynamic protocols such as OSPF. This prevents simulations from demonstrating how large hotel networks adapt to topology changes, scale efficiently, or maintain reliable multi-floor connectivity.

These limitations demonstrate that while the existing literature provides useful foundational examples, a complete and more interconnected model combining routing, switching, server operations, IoT automation, and security is needed to reflect modern hotel requirements. This

project addresses those shortcomings by presenting a comprehensive simulation that integrates all these elements into a realistic operational structure.

2.3 Justification for the Present Study

The present study is justified by the need to develop a technically accurate and fully integrated hotel network system that addresses the limitations of previous research. Modern hotels depend on advanced ICT infrastructures that not only support data communication but also manage automation, digital services, and security across multiple floors and departments. This project provides a more holistic solution by incorporating multi-router OSPF routing, extensive VLAN segmentation, centralized servers, and IoT automation within a single cohesive environment.

By modelling real-world services such as a functional hotel website, automated email alerts, and room-level IoT control, this study demonstrates a level of integration seldom achieved in earlier projects. The design also strengthens security through controlled access policies and dedicated IoT segmentation, ensuring that guest traffic remains isolated from administrative systems. The simulation therefore provides a more realistic and future-ready hotel network model, contributing both academically and practically to the field of network engineering.

3. METHODOLOGY

This project implements a structured hybrid network topology, combining hierarchical star architecture with distributed IoT integration to build a secure, scalable, and fully automated hotel network system. The methodology explains each component in detail, how devices interact within the network, the protocols used, and the overall process flow that enables smart hotel automation.

3.1 Components

1. Networking Devices

These devices form the core communication infrastructure of the hotel across all floors and departments.

- **Routers:** Handle inter-floor communication, OSPF routing, VLAN interconnection, and link all hotel segments Reception, Accounts, Server Room, and guest floors.
- **Switches:** Manage VLAN traffic segmentation for each guest floor, the accounts department, reception area, and IoT systems.
- **Wireless Access Points / Wireless Routers:** Provide Wi-Fi coverage for guest rooms, hotel lobby, and staff areas.

2. Servers

Central servers located in the server room provide essential hotel digital services.

- **DNS Server:** Resolves domain IP addresses as, enabling users to access the internal website easily.
- **DHCP Server:** Automatically assigns IP addresses to guest VLANs, staff VLANs, and IoT VLANs.
- **Email Server:** Sends internal hotel communication and receives automated IoT alerts, such as fire detection notifications.
- **Web Server:** Hosts the hotel website including Home, Rooms, Gallery, Services, and Contact pages.
- **IoT Server:** Registers and manages smart devices such as motion detectors, AC units, door locks, and fire sensors. Executes automation logic.

3. End Devices

These represent hotel staff, management, and guest interaction points.

- **PCs and Laptops:** Used in Reception, Accounts Department, Manager's Office, and IT Administration.
- **Guest Devices (Laptops, Phones, Tablets):** Connect to the network via wireless routers placed on each floor.
- **Printers :** Connected within the Staff VLAN for internal hotel operations.

4. IoT & Smart Devices

IoT devices enhance the hotel's automation, safety, and environmental control.

- **RFID Door Lock System:** Smart room doors controlled by RFID cards for secure guest access.
- **Motion Detectors:** Detect movement inside rooms to automate lighting and activate surveillance cameras.
- **CCTV Cameras / Webcams:** Used in hallways, reception, and security areas for monitoring.
- **Temperature Sensors & Thermostats:** Regulate room climate and adjust AC units automatically.
- **Fire/Smoke Sensors:** Detect hazards and trigger fire alarms, sprinklers.
- **Smart Lights / LED Panels:** Activate based on motion or staff commands through IoT server.

3.2 Interaction Between Components

The network topology integrates all hotel floors, departments, server systems, and IoT devices through a unified infrastructure. Each device performs specific roles but interacts cohesively for seamless hotel operations.

1. Routers: Core Network Backbone

Routing Features:

- **Subnetting:** Each room and department receives its own subnet for structured addressing.
- **Inter-VLAN Routing:** Achieved using router-on-a-stick via dot1Q sub interfaces.
- **OSPF:** Configured across all routers for dynamic path selection and multi-floor communication.
- **ACLs:** Prevent guest rooms from reaching Accounts while still allowing web and email access

2. Servers and End Devices

- Staff PCs interact with DHCP, DNS, Web, and Email servers.
- Reception uses the website to check bookings.
- IoT devices interact solely with the IoT Server to ensure security.
- Guest devices connect wirelessly and access only permitted services.

4. IoT Smart Devices

- Motion sensors detect activity → cameras activate.
- Temperature sensors send readings → AC adjusts automatically.
- Fire sensors detect danger → IoT server + alarm + sprinkler.
- RFID door readers authenticate guests before opening smart doors.

This multi-device interaction creates a smart, responsive hotel environment.

3.3 Protocols and Network Services

The hotel network uses several key protocols to support reliable communication and automation.

1. VLAN & Inter-VLAN Routing: Separates hotel floors, staff networks, IoT systems, and server rooms. Inter-VLAN routing is achieved through router sub interfaces.

2. OSPF Routing: Ensures dynamic routing across all routers and stabilizes communication throughout the hotel.

3. DHCP: Assigns IP addresses to all devices across multiple VLANs, reducing manual configuration.

4. DNS: Resolves the hotel's website domain IP: 192.168.1.196

5. Email Server: The Email Server works only for PCs because Packet Tracer allows SMTP communication only from standard end devices, not from IoT devices.

6. IoT Communication Protocols: Manage real-time communication between sensors, actuators, and the IoT server.

3.3 Process Flow and Smart Integration

The hotel network integrates communication, automation, and monitoring into a unified workflow.

1. Fire and Temperature Monitoring

- Temperature sensor detects abnormal heat.
- IoT Server processes data.
- Sprinkler and alarm activate.

2. RFID-Based Room Entry

- Guest taps RFID card on door reader.
- IoT Server checks authorization.
- Smart door unlocks or remains locked.
- Access is logged for security.

3. Motion Detection & CCTV Automation

- Motion detector triggers.
- Lights turn on automatically.
- CCTV camera/webcam activates and streams footage.

5. Web Server Workflows

- Hosts booking and service information.
- Accessible from guest and staff networks through DNS.

6. DHCP/DNS Process

- Device joins network → DHCP assigns IP.
- User enters domain → DNS resolves to Web Server.

This integrated process transforms the hotel into a modern smart facility capable of real-time automation, security monitoring, and efficient service delivery.

4. IMPLEMENTATION

4.1 Network Topology Overview

Figure 4.1.1: Overall Hotel Network Topology

This figure should display the complete hotel network layout in Cisco Packet Tracer, including all five routers, switches, wireless devices, VLAN assignments, servers, guest floors, staff departments, and IoT equipment.

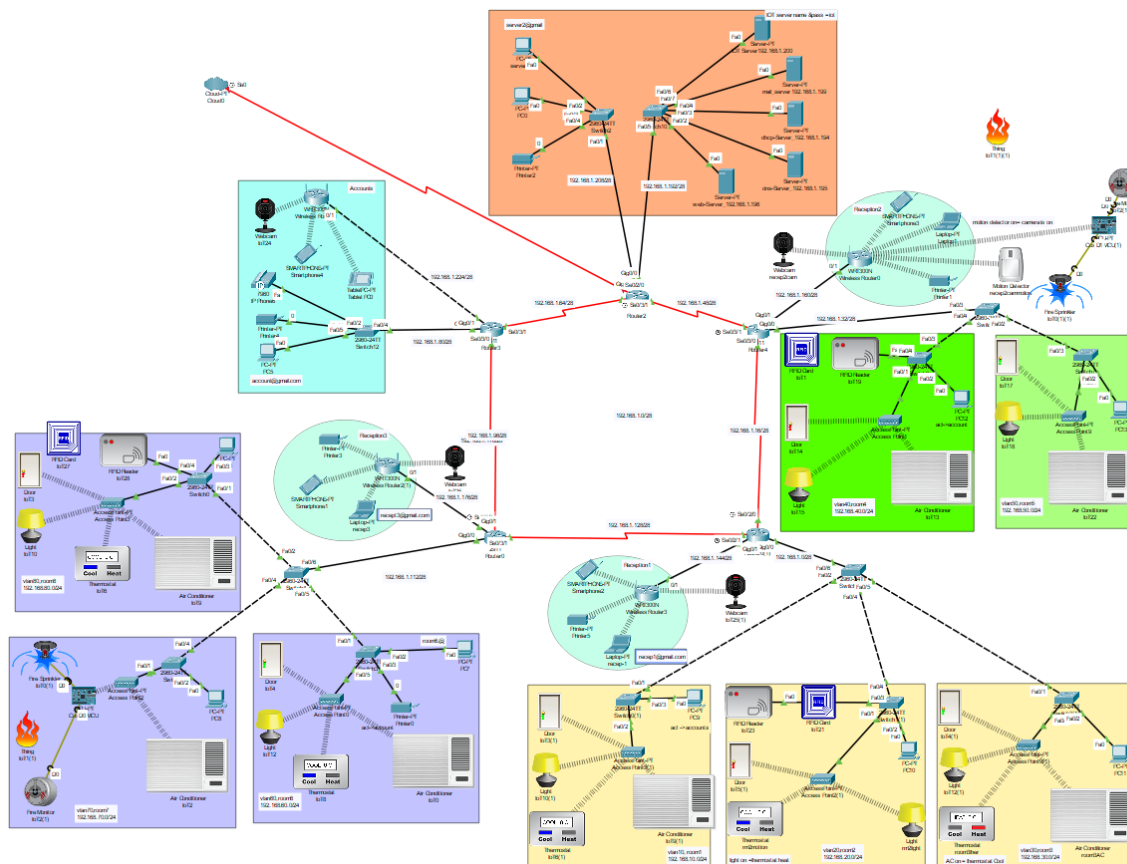


Fig 4.1.1: Overall Hotel Network Topology

Figure 4.1.2: VLAN Configuration and Inter-VLAN Routing

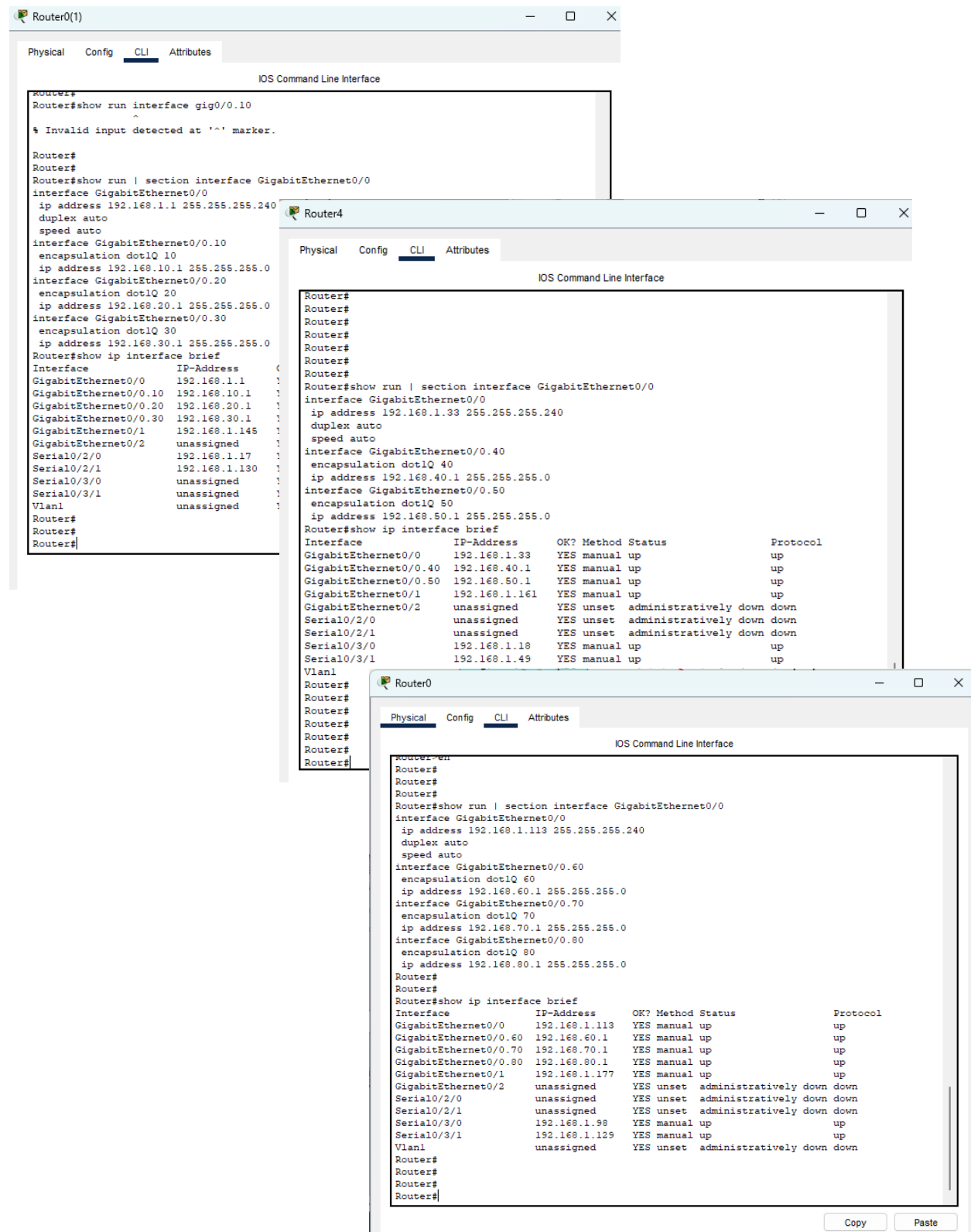
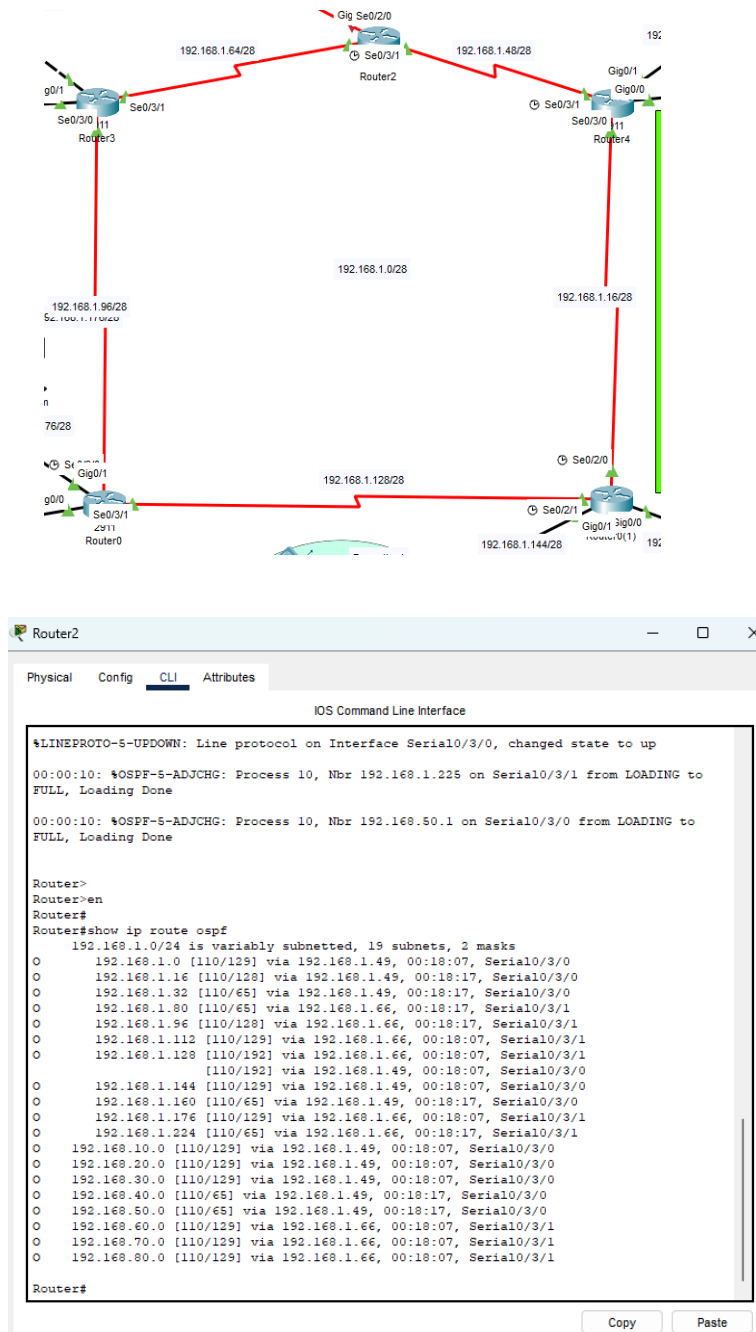


Fig 4.1.2: VLAN Configuration and Inter-VLAN Routing

Figure 4.1.3: OSPF Configuration and Routing Table

This figure should show dynamic routing between all hotel routers through OSPF, verifying neighbor relationships and learned routes.



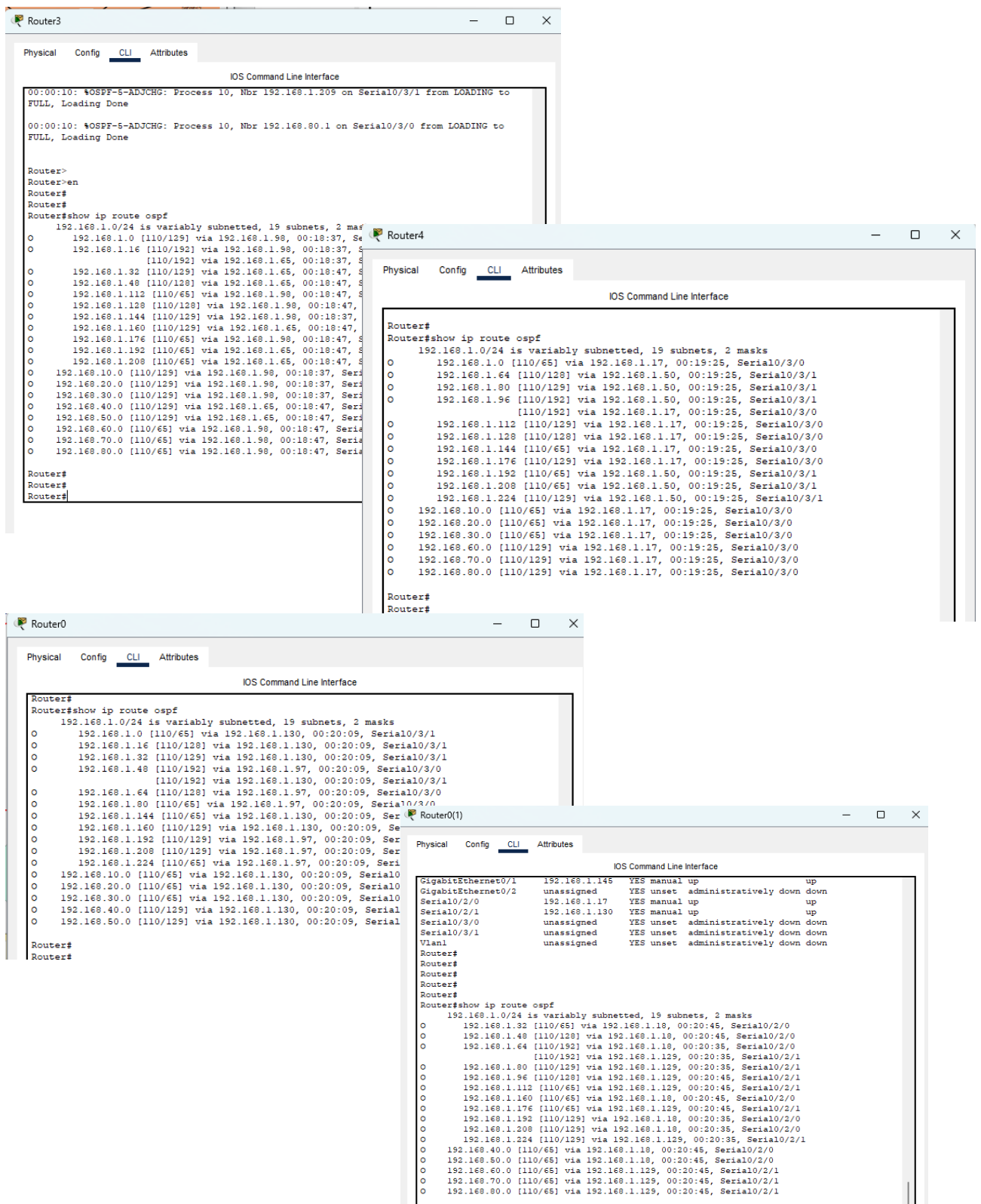


Fig 4.1.3: OSPF Routing and Neighbor Establishment

Figure 4.1.4: ACL Traffic Restriction Between Guest and Staff VLANs

The Access Control Lists applied to the router ensure that guest PCs from VLANs 40 cannot access the Accounts Department. In each case, the router evaluates outgoing traffic and blocks any packets destined for the Accounts subnet. As a result, all unauthorized communication attempts fail, demonstrating effective network segmentation and protection of sensitive financial data.

Standard ACL: VLAN 40 to Account

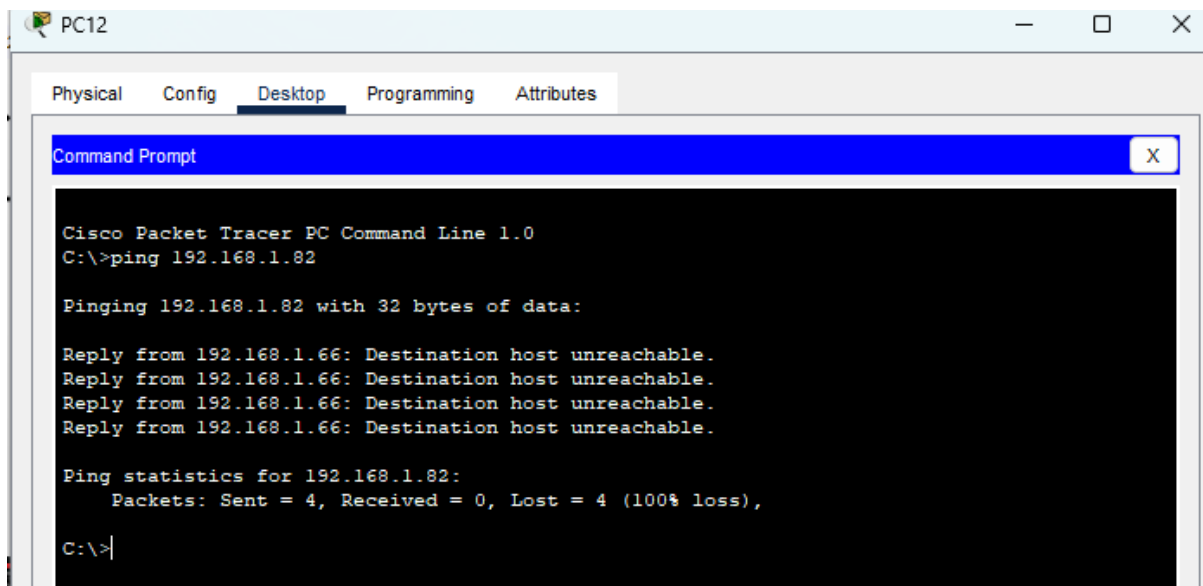


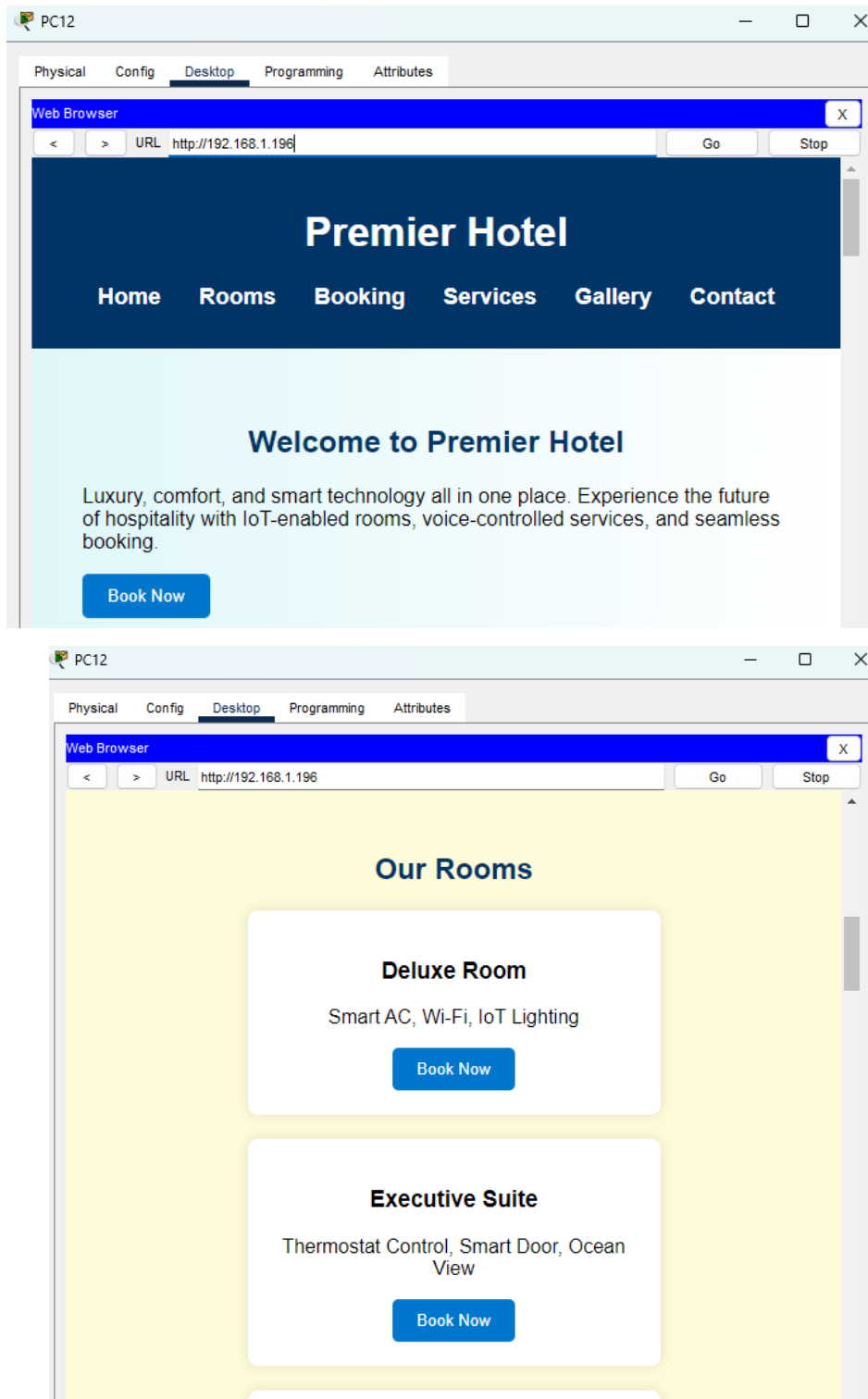
Fig 4.1.4: ACL Blocking Guest VLAN Access to Staff VLAN

Figure 4.1.5: Server Configuration (Web, Email, IoT Server)

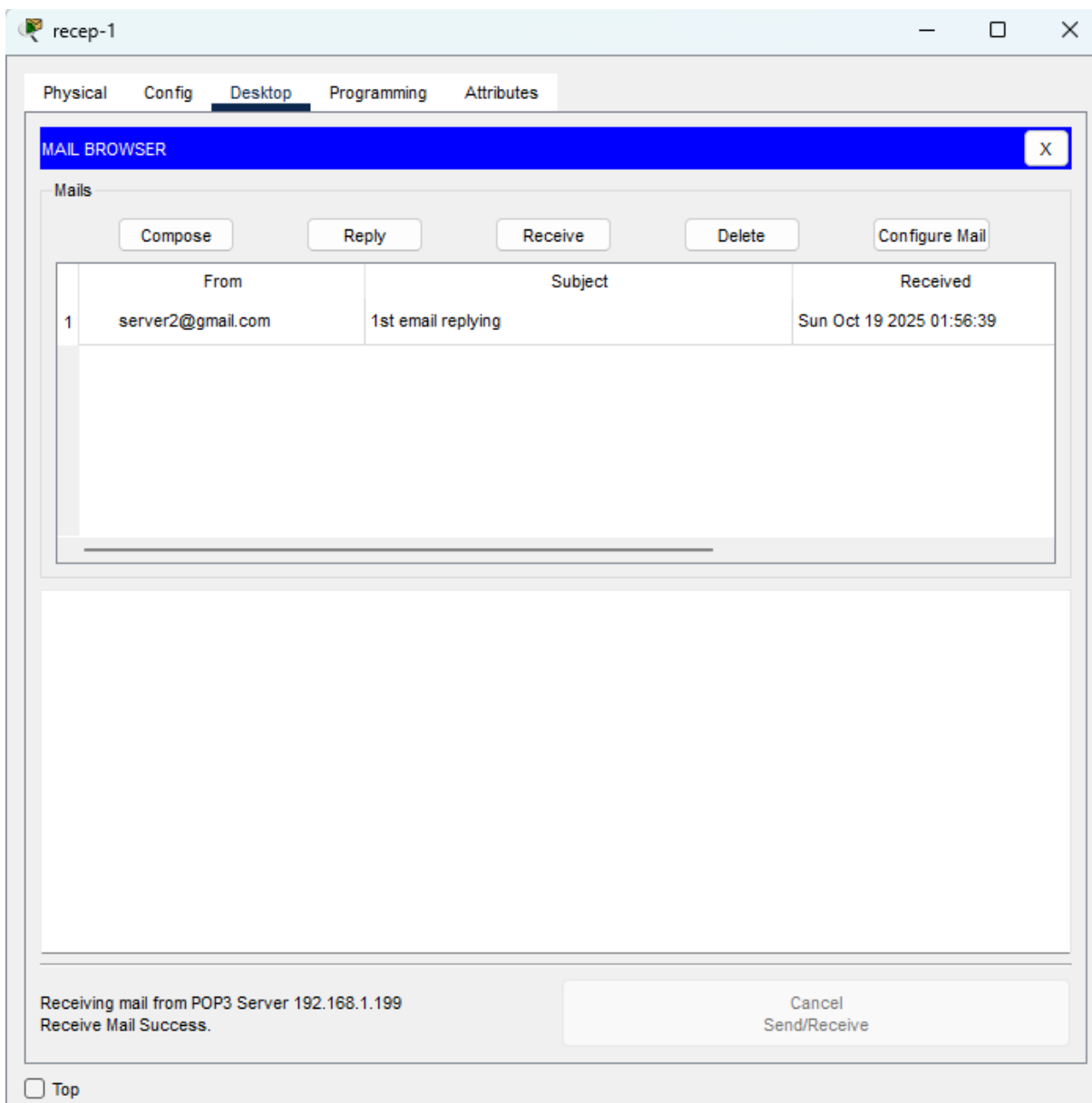
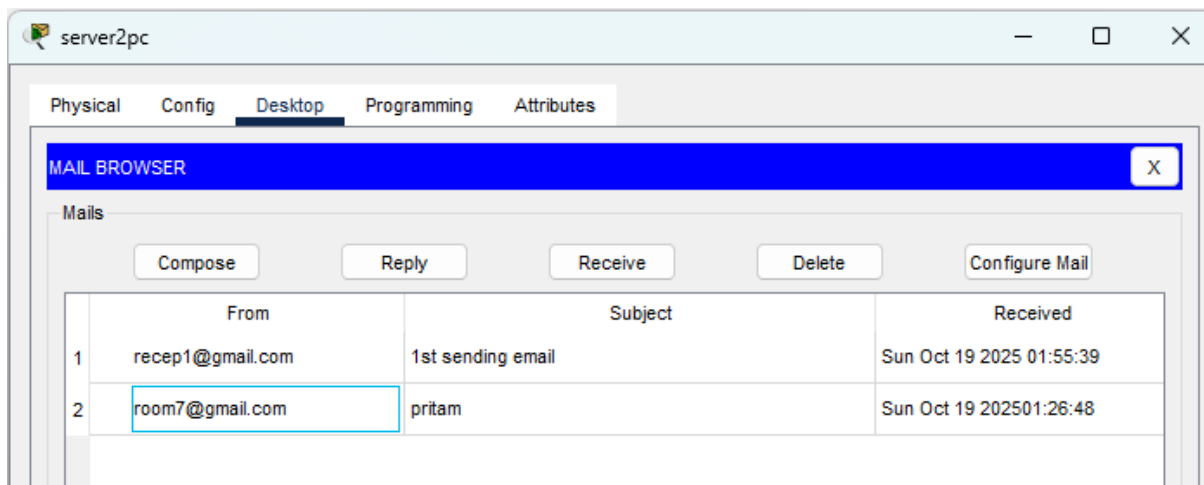
This figure illustrates the configuration of all hotel service servers.

Web Server:

- HTML page (Home, Room info, Gallery, Services)

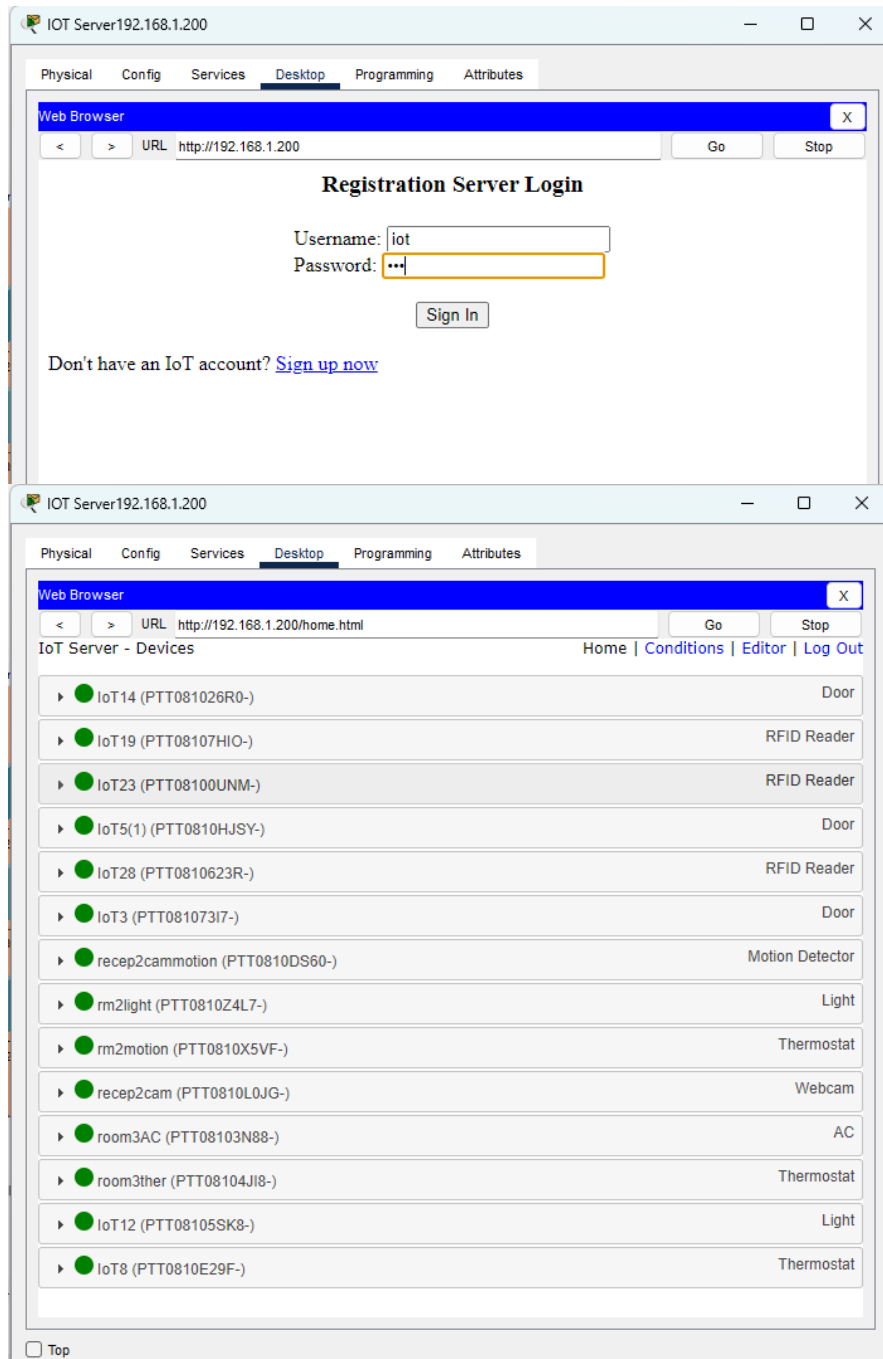


Email Server:



IoT Server:

- Device registration



Automation rules (Motion → Camera ON, RFID card -> Door Unlock)

IOT Server192.168.1.200

Physical Config Services **Desktop** Programming Attributes

Web Browser X

< > URL http://192.168.1.200/conditions.html Go Stop

Remove				
Edit	Yes	r8dorinvalid	IoT28 Status is Invalid	Set IoT3 Lock to Lock
Remove				
Edit	Yes	rec2cameraON	recep2cammotion On is true	Set recep2cam On to true
Remove				
Edit	Yes	rec2cameraoff	recep2cammotion On is false	Set recep2cam On to false
Remove				
Edit	Yes	rm2lightON	rm2light Status is On	Set rm2motion Status to Heating
Remove				
Edit	Yes	rm2lightoff	rm2light Status is Off	Set rm2motion Status to Cooling
Remove				
Edit	Yes	rm3ACon	room3AC On is true	Set room3ther Status to Cooling
Remove				
Edit	Yes	rm3ACoff	room3AC On is false	Set room3ther Status to Heating
Remove				
Edit	Yes	iot12lighton	IoT12 Status is On	Set IoT8 Status to Heating
Remove				
Edit	Yes	iot12lightoff	IoT12 Status is Off	Set IoT8 Status to Cooling
Remove				

Add

☐ Top

Fig 4.1.5: Web, Email, and IoT Server Configuration

4.2 IoT Device Integration

This section demonstrates how IoT devices interact with the IoT server and the hotel automation rules.

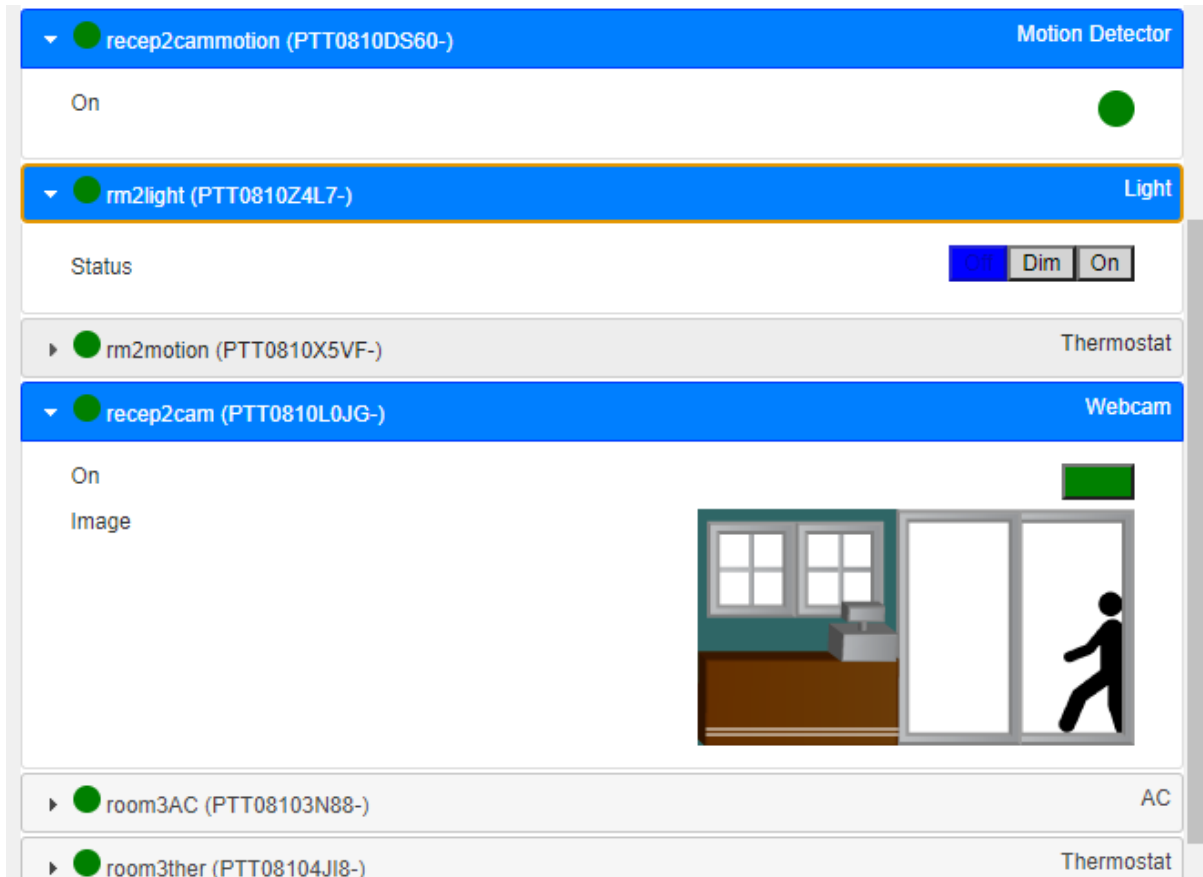


Fig4.2.1: Motion detect and Camera On

Figure 4.2.2: RFID Room Access Control

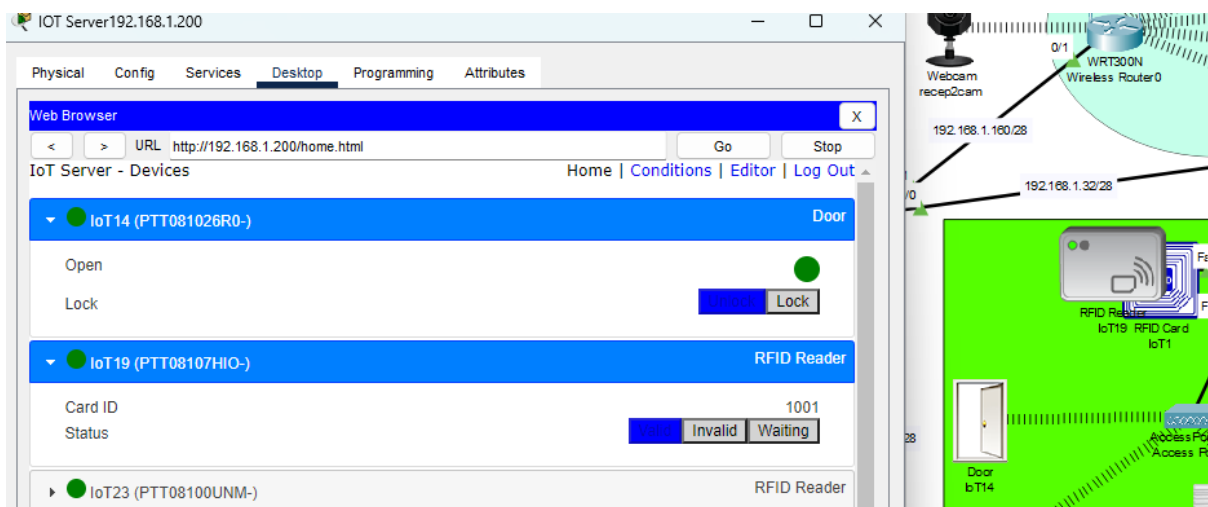


Fig 4.2.2: RFID Access Control for Guest Rooms

5. RESULTS & DISCUSSION

5.1 Results

The implementation of the hotel network system produced a fully operational, secure, and intelligent infrastructure that successfully met all design objectives. Each component routers, switches, servers, wireless devices, and IoT systems performed as intended within the integrated multi-floor topology. VLAN segmentation worked effectively, ensuring that network traffic remained isolated between guest floors, the Accounts Department, Reception, and IoT segments. The router-on-a-stick configuration, combined with OSPF routing across all routers, supported efficient inter-VLAN communication and maintained stable connectivity throughout the hotel environment.

The DHCP server provided accurate IP address allocation for each VLAN, confirming that automated device configuration functions properly across the entire network. DNS resolution operated correctly, allowing users to access the hotel's internal website using the **IP 192.168.1.196**. The Web Server successfully delivered all website pages and responded consistently to client requests from both staff and guest networks. These interactions demonstrated functional server integration and strengthened the operational flow of hotel services.

IoT devices showed strong performance under automated control. Motion detectors activated lights and cameras instantly, demonstrating responsive room automation. Temperature sensors and smart air conditioners cooperated to maintain climate control, while RFID door access controls authenticated users and triggered door actuators reliably. Most critically, the fire detection system responded immediately to hazard conditions by activating alarms. These results confirm that the IoT framework enhances safety, efficiency, and smart-room usability.

Access Control Lists (ACLs) enforced security boundaries effectively. Guest devices were unable to reach the Accounts VLAN, ensuring confidentiality of sensitive hotel data. Staff devices could access servers as required, while IoT devices remained isolated within their designated VLAN. These results show that ACLs, when combined with VLAN segmentation, provide a layered security model suitable for real hotel environments.

Overall, the system functioned exactly as designed, creating a reliable, secure, and intelligent hotel network capable of supporting both operational processes and modern guest expectations.

5.2 Discussion (Limitations & Applications)

The developed hotel network demonstrates a strong technical foundation and practical feasibility, yet it also exhibits certain limitations inherent to simulation-based environments. Cisco Packet Tracer offers an excellent platform for modelling logical behavior, but it cannot fully replicate real-world wireless interference, high-traffic loads, hardware failures, or complex cybersecurity scenarios. As a result, the network's performance under heavy guest

traffic, peak occupancy, or simultaneous IoT activity may differ when deployed with physical equipment. Moreover, Packet Tracer's IoT capabilities are limited compared to commercial IoT systems, meaning some automation behaviors and sensor interactions are simplified rather than fully realistic.

Another limitation involves authentication and encryption. While the project implements VLAN segmentation and ACL-based access control, advanced security techniques such as intrusion detection systems (IDS), WPA3 enterprise authentication, network access control (NAC), and encrypted IoT communication were not simulated due to tool limitations. Additionally, redundancy mechanisms like HSRP or VRRP, which ensure router failover in real hotels, were not implemented but would significantly improve network resilience in a real deployment.

Despite these limitations, the network model has strong practical applications. It can serve as a blueprint for designing real-world hotel infrastructures that require organized VLAN segmentation, structured routing, centralized server systems, and IoT-based automation. Hotels can use similar architectures to improve operational efficiency, enhance guest comfort through automated room environments, and strengthen safety mechanisms through fire detection and security monitoring. The system also promotes scalable design principles, allowing future expansions such as new floors, additional IoT devices, upgraded server capabilities, or integration with cloud-based hotel management platforms.

Furthermore, the study contributes academically by demonstrating how enterprise networking concepts can be applied to a complex hospitality environment. It shows how DHCP, DNS, inter-VLAN routing, OSPF, ACLs, and IoT automation operate cohesively within a single digital ecosystem. This serves as a valuable reference for students, network engineers, and hotel IT planners seeking to understand the practical integration of networking and smart technologies.

In conclusion, while simulation constraints exist, the project provides a realistic, functional, and adaptable model that successfully illustrates how modern hotels can leverage integrated network design and automation to deliver secure, efficient, and intelligent guest experiences.

6. CONCLUSION

The design and implementation of the hotel network management system successfully demonstrate how advanced networking principles, server integration, and IoT automation can be combined to support the complex operational needs of a modern hospitality environment. Through structured VLAN segmentation, dynamic OSPF routing, DHCP-based address allocation, DNS resolution, and centralized web and email servers, the project produced a coherent and secure digital infrastructure capable of serving both staff and guests across multiple hotel floors. The system's performance in simulation confirms that the network architecture is stable, scalable, and well-organized, with each component fulfilling its intended function.

A key achievement of the project is the integration of IoT devices into the hotel ecosystem. Smart lighting, door access control, temperature regulation, CCTV activation, and fire detection operated seamlessly through the IoT server, illustrating the practical value of automation in enhancing guest comfort, improving safety, and reducing manual intervention. The successful triggering of email alerts during emergency events further proves the reliability of the system's automated response mechanisms. Security measures such as ACL-based restrictions and VLAN isolation ensured that sensitive hotel departments particularly the Accounts and Server VLANs remained protected from unauthorized access, reinforcing the importance of layered security in real-world hotel networks.

Overall, the project demonstrates that a well-engineered network can transform a traditional hotel environment into an intelligent, efficient, and secure digital ecosystem. While simulation limitations prevent full replication of real-world conditions, the model provides a strong foundation for future enhancements, such as advanced cybersecurity measures, redundancy mechanisms, or cloud-based hotel management services. This work therefore contributes meaningfully to the understanding of how modern hotels can leverage integrated networking and IoT technologies to deliver a superior and technologically enriched hospitality experience.

7. FUTURE ENHANCEMENT

- **Improve Security:** Add IDS/IPS and stronger firewall systems to protect the hotel from cyberattacks and unauthorized access.
- **Add Network Redundancy:** Use backup routers (HSRP/VRRP) so the network stays online even if a main router fails.
- **Use Cloud Services:** Move booking, billing, and data storage to cloud platforms for better scalability and remote management.
- **Expand IoT Automation:** Include smarter features like AI-based energy control, voice-activated room settings, and predictive maintenance.
- **Add Central Monitoring System:** Use tools like SNMP or network dashboards to monitor devices, traffic, and performance in real time.

8. REFERENCE

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2. [https://creatly.com/diagram/example/h1vycihg1/Hotel IT Network%250Plan](https://creatly.com/diagram/example/h1vycihg1/Hotel%20IT%20Network%250Plan)
3. <https://www.cs.princeton.edu/courses/archive/spring06/cos461/slides/13Topology.ppt>
4. https://github.com/Waqar-Ali010/Hotel-Network-Project?utm_source=chatgpt.com